Project report

1. Dataset and Simulation Samples

This analysis uses proton–proton collision data collected by the CMS experiment at a center-of-mass energy of 13 TeV during the 2016 LHC Run (Run II, Ultra Legacy dataset). The data correspond to the DoubleEG trigger stream, stored in the NANOAOD format (NanoAODv9).

The real data sample:

- 2016 Data (Run H, DoubleEG dataset) CMS Run II Ultra Legacy (2016H), reconstructed in MiniAODv2 and processed to NanoAODv9 format. Dataset: CMS Open Data record 30554

Monet Carlo simulation samples

Signal (Drell-Yan)

- DYJetsToLL (M-50)
  - Dataset: CMS Open Data record 35669
- DYJetsToLL (M-10 to 50)
  - Dataset: CMS Open Data record 35631

Background processes

The following Standard Model background processes are included:

- **Top quark pair production ($t\bar{t}$)**
  - Dataset: CMS Open Data record 67731
- **Single top production**
  - tW channel (top + W)
    - Dataset: CMS Open Data record 64895
  - t{W_bar} (anti-top W)
    - Dataset: CMS Open Data record 64839
  - t-channel single top
    - Dataset: CMS Open Data record 64793
  - t-channel single anti-top
    - Dataset: CMS Open Data record 64693
- **Diboson processes**
  - WW
    - Dataset: CMS Open Data record 72696
  - WZ
    - Dataset: CMS Open Data record 72754
  - ZZ
    - Dataset: CMS Open Data record 75593
- **Drell-Yan** $\rightarrow \tau\tau$

– Events where $Z/\gamma^* \to \tau^+\tau^-$ are treated as a separate background. These are separated using LHE-level tau identification (PDG ID = ± 15).

2. Event Selection and Feature Construction (Data Preparation)

Feature Engineering and Event Selection

In this work, a physics-driven feature preparation pipeline was implemented using ROOT-based analysis of NanoAOD data. The goal was to construct fixed-size, machine-learning-ready event representations for a neural network classifier.

Event Topology Selection

Events are required to contain at least two electrons, which are then sorted by transverse momentum and reduced to a fixed two-electron system (leading and subleading electrons). This defines a dilepton event topology, consistent with processes such as Drell–Yan (DY $\to e^+e^-$).

Kinematic and Acceptance Cuts

A set of standard CMS-inspired selection criteria is applied: - Leading electron: $p_T > 28$ GeV - Subleading electron: $p_T > 20$ GeV - Pseudorapidity: $|\eta| < 2.5$ - ECAL barrel–endcap transition region removed: $1.4442 < |\eta| < 1.566$

These cuts ensure detector acceptance, trigger compatibility, and improved object quality.

Physics-Inspired Feature Set

Each event is transformed into a fixed-length feature vector consisting of:

Electron Features (2 objects per event)

- p_T, η, ϕ
- Shower shape: `sieie`, `hoe`
- Impact parameters: `dxy`, `dz`
- Isolation variables: `miniPFRelIso_all`, `dr03TkSumPt`
- Energy consistency: `eInvMinusPInv`, `scEtOverPt`

Jet Features (up to 4 jets)

- p_T, η, ϕ
- b-tag discriminator `btagDeepFlavB` #### Photon Features (up to 2 photons)
- p_T, η, ϕ
- Shower shape: `sieie`, `hoe`

- Isolation `pfRelIso03_all` Photons are matched to electrons using a minimum ΔR criterion to capture nearby radiation or misidentified objects.

Missing Transverse Energy (MET)

- `MET_phi`
- `MET_significance`
- `MET_sumEt`

3. Neural Network Architecture and Training

A fully connected feed-forward neural network was used to perform binary classification between signal (Drell–Yan) and Standard Model background processes.

Input Features

The model is trained on a fixed-length feature vector constructed from: - Electron kinematics and identification variables - Jet activity (up to 4 jets) - Photon activity (up to 2 photons) - Missing transverse energy (MET) variables

Exact feature names were mentioned above. All input features are standardized using StandardScaler to ensure zero mean and unit variance, improving training stability.

Model Architecture

The neural network is implemented using TensorFlow/Keras and has the following structure: - Input layer: number of physics features - Dense layer: 48 neurons - Batch Normalization - LeakyReLU activation - Dropout (0.2) - L2 regularization (1e-4) - Dense layer: 24 neurons - Batch Normalization - LeakyReLU activation - Dropout (0.2) - Dense layer: 12 neurons - LeakyReLU activation - Output layer: - 1 neuron with sigmoid activation - Outputs probability of being signal (DY) vs background

Training Setup

The model is trained as a binary classifier using: - Loss function: binary cross-entropy - Optimizer: Adam (learning rate = 5×10^{-4}) - Metrics: accuracy and ROC-AUC - Batch size: 128 - Epochs: 30 - Validation split: 20% of training data

Sample Weighting Strategy

To improve physical realism and reduce bias in the training distribution, a custom sample weighting strategy is applied.

1. Signal reweighting (mass flattening) Signal events are reweighted based on the invariant mass distribution of the reconstructed Z boson system. The mass spectrum is binned, and weights are assigned as: - Events in bin i are weighted by:

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$w = \frac{1}{\sqrt{N_{\text{bin}}}}$
```

where: - N_{bin} = number of events in mass bin.

This flattens the mass distribution and prevents the model from being dominated by high-statistics regions.

2. Background treatment and QCD enhancement All background processes are included in training, with special treatment for QCD: - QCD samples (qcd1–qcd8) are explicitly upweighted ($\times 10$) - This improves sensitivity to rare but important QCD-like backgrounds - Ensures the classifier does not underestimate QCD contamination

Dataset Splitting and Evaluation Strategy

The dataset is randomly split into training (80%) and testing (20%) subsets. Additionally, independent Monte Carlo samples not used in training are used for evaluation of model performance, ensuring unbiased validation and preventing data leakage. ### Output The trained model outputs a probability score: - Close to 1 $\rightarrow$ signal-like event (DY) - Close to 0 $\rightarrow$ background-like event The final model is saved for further physics analysis and inference studies.

Results and Discussion

To evaluate the performance of the neural network classifier, its results are compared against a traditional cut-based selection implemented in C++. The comparison is performed using two key physics performance metrics: - **Efficiency per bin:**

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Efficiency is defined as the fraction of true signal events that pass a given selection:

$Efficiency = \frac{N_{signal, \ selected}}{N_{signal, \ total}}$


It measures how well the selection retains genuine signal events. High efficiency means the
```

- **Purity per bin:**

  Purity is defined as the fraction of selected events that are true signal:

  $Efficiency = \frac{N_{signal,\ selected}}{N_{signal,\ selected} + N_{background,\ selected}}$

  It measures how clean the selected sample is in terms of signal content. High purity means low background contamination.

4.1 Baseline: Pre-Selection (Before Classification) Before applying any selection or machine learning model, the dataset shows:

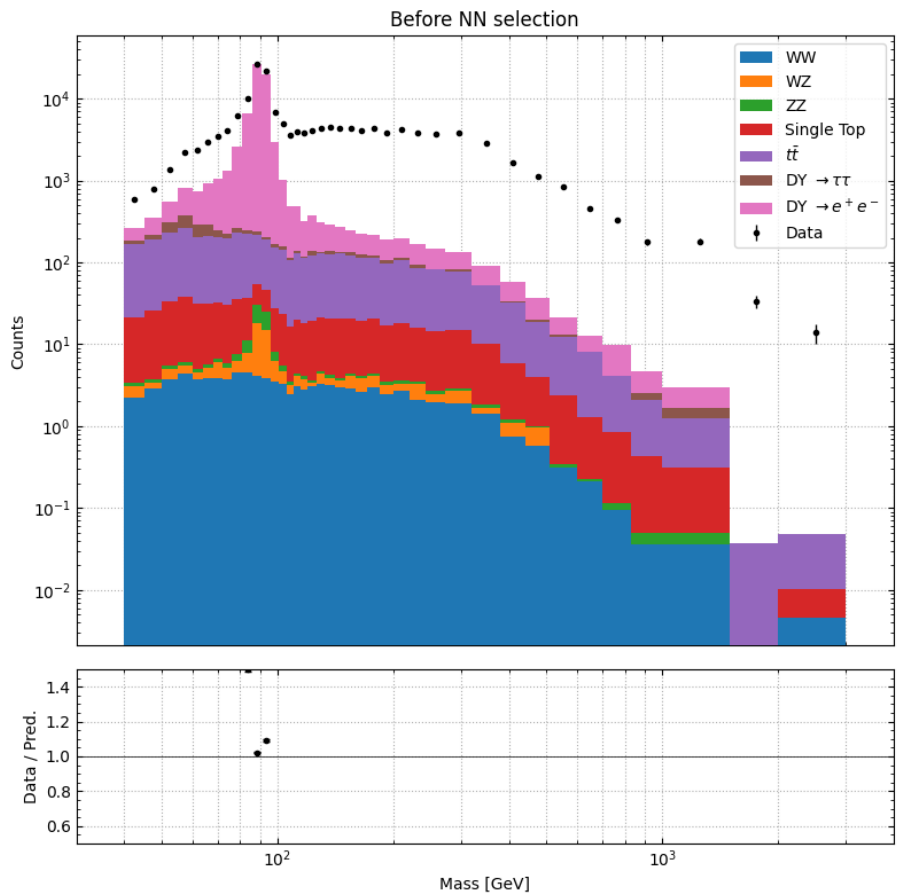

Figure 1: initial

This is reference distribution for both methods.

4.2 Cut-Based Selection (C++ Analysis) A traditional physics-driven selection is applied using fixed kinematic and object-level cuts implemented in C++. The plot is shown below:

Event Selection Requirements

Events are required to satisfy: - Double-electron trigger - `HLT_Ele23_Ele12_CaloIdL_TrackIdL_IsoVL_DZ` - At least two reconstructed electrons - Electron identification - Medium cut-based electron ID (cutBased ≥ 3) - Isolation requirement - Relative isolation: `pfRelIso03_all < 0.15` - Opposite-sign electron pair - Selected electrons must

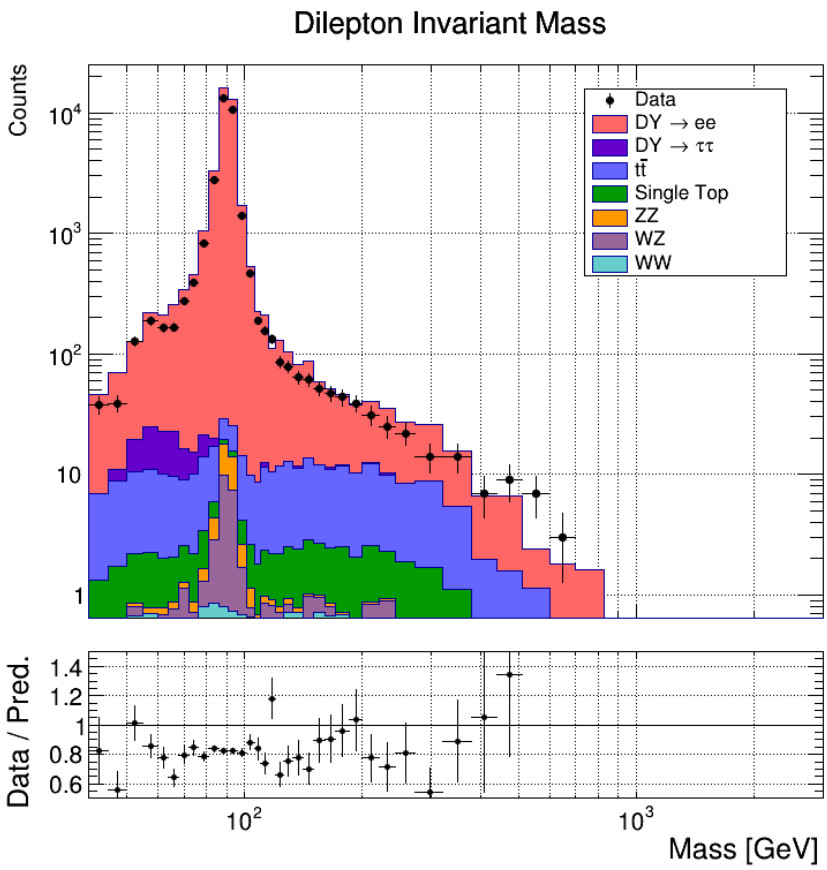
Dilepton Invariant Mass
Counts
Data
DY → ee
DY → ττ
tt̄
Single Top
ZZ
WZ
WW
Data / Pred.
Mass [GeV]

Figure 2: cpp

satisfy opposite electric charge - Transverse momentum requirements - Leading electron: $p_T > 28$GeV - Subleading electron: $p_T > 20$GeV - Detector acceptance - $|\eta| < 2.5$ - ECAL transition region veto - Events are rejected if either electron lies in $1.4442 < |\eta| < 1.566$

to avoid reduced detector response in the barrel–endcap transition region.

Results:

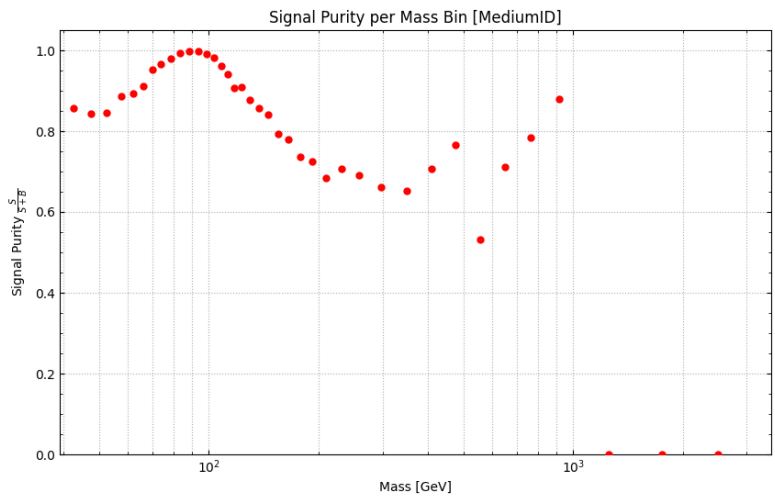

Figure 3: cpp purity

4.3 Neural Network Classification Results The neural network classifier significantly improves event separation. The plot is shown below:

Results:

Key observations: - Clear improvement in signal-to-background discrimination - Higher purity across most invariant mass bins compared to cut-based selection - Higher efficiency than when using cut-based selection: less signal is dropped out.

4.4 Efficiency and Purity Comparison The comparison between the two methods shows: - Efficiency per bin: - Neural network maintains higher signal efficiency - Cut-based selection shows sharper drops in certain regions due to hard thresholds - Purity per bin: - Neural network achieves consistently higher and uniform purity across most bins - Especially improved separation in regions with high background overlap (low and intermediate mass ranges)

Overall, the neural network provides a more optimal balance between efficiency and purity.

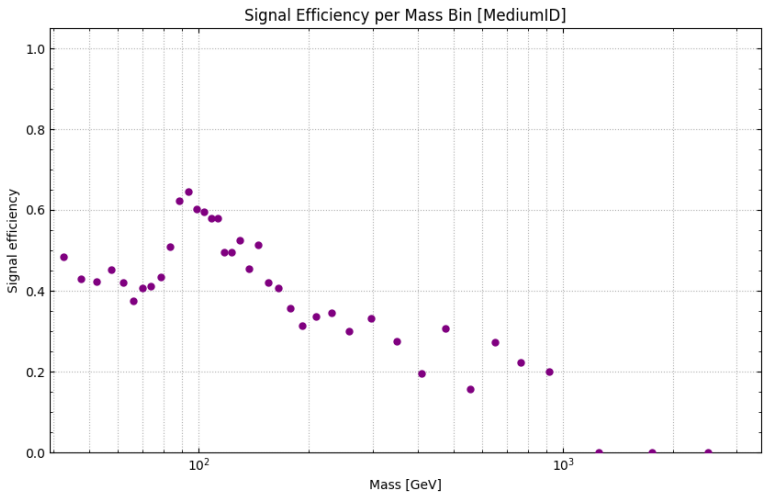

Figure 4: cpp efficiency

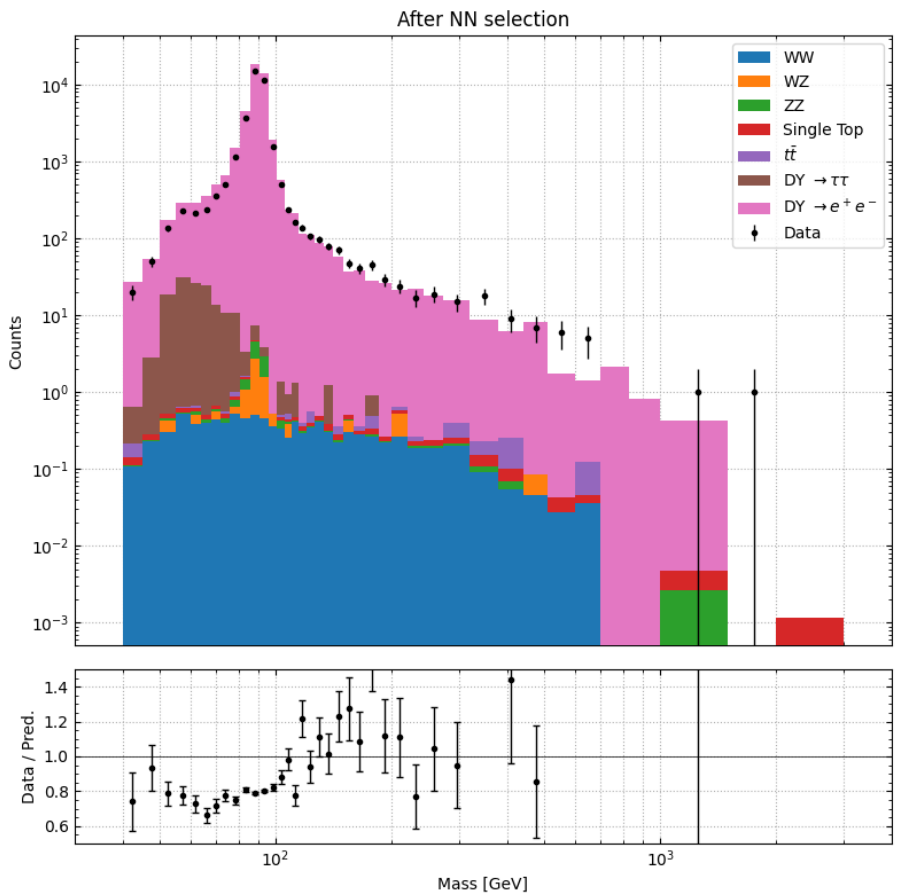
After NN selection
WW
WZ
ZZ
Single Top
$t\bar{t}$
DY $\to \tau\tau$
DY $\to e^+e^-$
Data
Counts
Data / Pred.
Mass [GeV]

Figure 5: nn

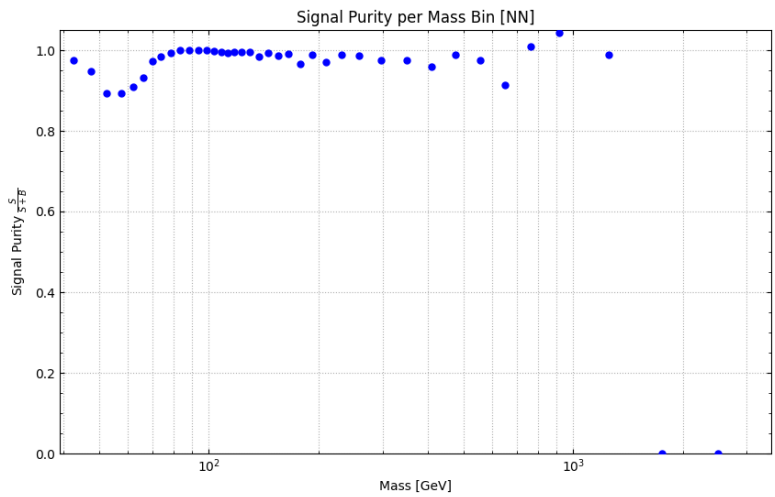

Figure 6: nn purity

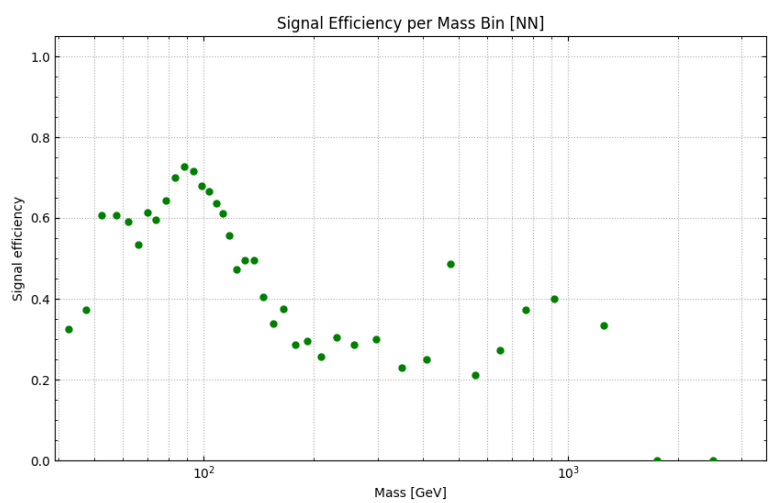

Figure 7: nn efficiency